F18CE2 - MV Calculus and Real Analysis B: Solutions 9

1

$$L(P,f)=\sum_{r=0}^{k-1} I_r(f)(x_{r+1}-x_r), \quad U(P,f)=\sum_{r=0}^{k-1} S_r(f)(x_{r+1}-x_r),$$
$$I_r(f)=\inf\{f(x): x\in[x_r,x_{r+1}]\}, \quad S_r(f)=\sup\{f(x): x\in[x_r,x_{r+1}]\}.$$

2

$$U(P,f)=9(1)^2\left(\frac{2}{3}-0\right)+9\left(\frac{1}{3}\right)^2\left(\frac{4}{3}-\frac{2}{3}\right)+9(1)^2\left(2-\frac{4}{3}\right)=\frac{38}{3}.$$

3 $U(P,f)=3\,S(f)$ where $S(f)=\sup\{f(x): 0\leqslant x\leqslant 3\}$.

$U(Q,f)=2\,S_1(f)+S_2(f)$ where

$$S_1(f)=\sup\{f(x): 0\leqslant x\leqslant 2\}, \quad S_2(f)=\sup\{f(x): 2\leqslant x\leqslant 3\}.$$

Since $S_r(f)\leqslant S(f)$ for $r=1,2$,

$$U(Q,f)\leqslant 2\,S(f)+S(f)=3\,S(f)=U(P,f).$$

4

$$U(P_n,f)=0\times\left(\frac{1}{2}-\frac{1}{n}\right)+5\times\frac{2}{n}+0\times\left(1-\frac{1}{2}-\frac{1}{n}\right)=\frac{10}{n},$$

and $L(P_n,f)=0$. Since

$$\lim_{n\to\infty}(U(P_n,f)-L(P_n,f))=\lim_{n\to\infty}\frac{10}{n}=0.$$

so f is integrable on $[0,1]$ and $\displaystyle\int_0^1 f(x)\,dx=\lim_{n\to\infty}U(P_n,f)=0.$

5

$$U(P_n,f)=-2\left(6-\frac{1}{n}\right)+5\left(6+\frac{1}{n}-\left(6-\frac{1}{n}\right)\right)+4\left(10-\left(6+\frac{1}{n}\right)\right)=4+\frac{8}{n}\cdot$$
$$L(P_n,f)=-2\left(6-\frac{1}{n}\right)+-2\left(6+\frac{1}{n}-\left(6-\frac{1}{n}\right)\right)+4\left(10-\left(6+\frac{1}{n}\right)\right)=4-\frac{6}{n}\cdot$$

$$\lim_{n\to\infty}(U(P_n,f)-L(P_n,f))=\lim_{n\to\infty}\frac{14}{n}=0.$$

Hence f is integrable on $[0,10]$ and

$$\int_0^{10} f(x)\,dx=\lim_{n\to\infty}U(P_n,f)=\lim_{n\to\infty}4+\frac{8}{n}=4.$$

6 Since f is increasing, for each $r = 1, \ldots, k-1$,

$$S_r(f) = \sup\{f(x) : x \in [x_r, x_{r+1}]\} = f(x_{r+1}), \quad I_r(f) = \inf\{f(x) : x \in [x_r, x_{r+1}]\} = f(x_r),$$

so

$$U(P,f) = \sum_{r=0}^{k-1} f(x_{r+1})\,(x_{r+1} - x_r)\,, \quad L(P,f) = \sum_{r=0}^{k-1} f(x_r)\,(x_{r+1} - x_r)\,.$$

7 Since $I_r(f) \geqslant m$

$$L(P,f) = \sum_{r=0}^{k-1} I_r(f)(x_{r+1} - x_r) \geqslant \sum_{r=0}^{k-1} m(x_{r+1} - x_r) = m\sum_{r=0}^{k-1}(x_{r+1} - x_r) = m(b-a),$$

$$\int_a^b f(x)dx = \sup\{L(P,f) : P \text{ a partition of } [a,b]\} \geqslant m(b-a).$$

For $a \leqslant x \leqslant a+1$,

$$f(x) = \frac{1}{x} \geqslant \frac{1}{a+1}\,.$$

Hence,

$$\int_a^{a+1} \frac{1}{x}\,dx = \ln(1+a) - \ln a = \ln(1+a^{-1}) \geqslant \frac{1}{a+1}.$$

8 Using the partition suggested in the question:

(a) $$L(P_n,f) = \sum_{r=0}^{n-1} I_r(f)(x_{r+1} - x_r) = \sum_{r=0}^{n-1} \frac{r}{n}\frac{1}{n} = \frac{1}{n^2}\sum_{r=0}^{n-1} r = \frac{1}{n^2}\frac{n(n-1)}{2} \to \frac{1}{2}.$$

We could also work out the upper sums to show that also $U(P_n,f) \to \frac{1}{2}$, and so $f(x) = x$ is integrable and the integral is $\frac{1}{2}$.

However, we know that the function $f(x) = x$ is continuous, so is integrable, and by Corollary **??** in the notes, just working out the above limit is sufficient to give us the value of the integral.

By the fundamental theorem of calculus, $\displaystyle\int_0^1 x\,dx = [\tfrac{1}{2}x^2]_0^1 = \frac{1}{2}.$

(b) $$L(P_n,f) = \sum_{r=0}^{n-1} I_r(f)(x_{r+1} - x_r) = \sum_{r=0}^{n-1} \frac{r^2}{n^2}\frac{1}{n} = \frac{1}{n^3}\sum_{r=0}^{n-1} r^2 = \frac{1}{n^3}\frac{(n-1)n(2n-1)}{6} \to \frac{1}{3}.$$

As in part (a), this is enough to give us the value of the integral.

By the fundamental theorem of calculus, $\displaystyle\int_0^1 x^2\,dx = [\tfrac{1}{3}x^3]_0^1 = \frac{1}{3}.$

The fundamental theorem of calculus is obviously a much easier way of doing integration!

9 (a) By 'normal' integration on the separate intervals $[-1, 0]$ and $[0, 1]$ (using the Fundamental Theorem of Calculus), we see that

$$G(x) := \begin{cases} 0, & -1 \leqslant x \leqslant 0, \\ x, & 0 < x < 1. \end{cases}$$

It is easy to see that G is continuous on $[-1, 1]$, and differentiable everywhere on $(-1, 1)$ except at $x = 0$. At $x = 0$ we have

$$\lim_{h \to 0^-} \frac{G(h) - G(0)}{h} = 0, \quad \lim_{h \to 0^+} \frac{G(h) - G(0)}{h} = \lim_{h \to 0^+} \frac{h}{h} = 1,$$

and since these one-sided limits are not equal to each other, the limit does not exist, so G is not differentiable at $x = 0$.

(b) By similar calculations to previous examples, it is easy to see that $G(x) = 0$, for all $x \in [-1, 1]$, so obviously $G'(x) = 0$. Hence, at $x = 0$, $G'(0) \neq g(0)$.